P1

HW 03

$$V(x,y,z) = A\cos(x-2y) + B\sin(y+3z) + C\cos(x+6z)$$

$$T = \frac{1}{2}m(\dot{x}^2+\dot{y}^2+\dot{z}^2)$$

$\vec{r} \to \vec{r} + \xi\vec{a}$ where ξ = parameter

$\vec{a}$ = constant vector

$$\begin{cases} x \to x + a_1\xi \\ y \to y + a_2\xi \\ z \to z + a_3\xi \end{cases} \qquad \begin{cases} x-2y \to x-2y + \xi(a_1-2a_2) \\ y+3z \to y+3z + \xi(a_2+3a_3) \\ x+6z \to x+6z + \xi(a_1+6a_3) \end{cases}$$

Require $\begin{cases} a_1 = 2a_2 \\ a_2 = -3a_3 \\ a_1 = -6a_3 \end{cases} \Rightarrow$ let $a_3 = \alpha$

$\Downarrow$

$a_2 = -3\alpha$, $a_1 = -6\alpha$

$$\Rightarrow \begin{cases} x \to x - 6\alpha\xi \\ y \to y - 3\alpha\xi \\ z \to z + \alpha\xi \end{cases}$$

with arbitrary α leaves $V(x,y,z)$ invariant

"Conserved charge" $A = \dfrac{\partial L}{\partial \dot{q}_\sigma}\dfrac{\partial \hat{q}_\sigma}{\partial \xi}\Big|_{\xi=0}$ (Noether's Th)

where $q_\sigma \to \hat{q}_\sigma(q,\xi)$

$$A = m\dot{x}\frac{\partial\tilde{x}}{\partial\xi} + m\dot{y}\frac{\partial\tilde{y}}{\partial\xi} + m\dot{z}\frac{\partial\tilde{z}}{\partial\xi}\Big|_{\xi=0}$$

$$= \alpha\left[(-6)m\dot{x} + (-3)m\dot{y} + m\dot{z}\right] = (-\alpha)(6m\dot{x} + 3m\dot{y} - m\dot{z})$$

Conserved $A = (-\alpha)(6p_x + 3p_y - p_z)$.

Problem 2

(a)

As the charges Q_1 and Q_2 have infinite masses they do not move and the kinetic energy of the system is

$$T = \frac{m\dot{\vec{r}}^2}{2},$$

where $\vec{r}$ is the radius vector of the charge q, and m is its mass. Potential energy of the system is

$$U = \frac{kqQ_1}{|\vec{R}_1 - \vec{r}|} + \frac{kqQ_2}{|\vec{R}_2 - \vec{r}|}.$$

Thus the Lagrangian reads:

$$\mathcal{L} = \frac{m\dot{\vec{r}}^2}{2} - \frac{kqQ_1}{|\vec{R}_1 - \vec{r}|} - \frac{kqQ_2}{|\vec{R}_2 - \vec{r}|}.$$

(b)

The Lagrangian is invariant under the rotations of space around the axis passing through the two points $\vec{R}_1$ and $\vec{R}_2$. Consequently, the component of the angular momentum along this axis is conserved. There is no conserved component of the linear momentum.

Problem 3

The potential energy and the Lagrangian are invariant under the transformation

$$(\rho, \phi, z) \rightarrow (\rho, \phi, z) + \xi(0, 1, -\alpha)$$

As a result, the quantity,

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} - \alpha \frac{\partial \mathcal{L}}{\partial \dot{z}} = M_z - \alpha P_z$$

is an integral of motion. A charged pointlike particle in the field of charged infinitely long cylindrical helix would realize this kind of system. In this case α is proportional to the step of the helix.